How the Data We Make Can Unmake Us: Annals of Factology

Bruce L. Gardner

The conclusions that agricultural economists have been able to reach on almost every issue of substantive scientific interest depend on the data available. Moreover, the conclusions often depend upon the data themselves telling a story (rather than data recast or purified using econometrics). When the data speak for themselves we tend to listen and believe without the critical scrutiny we give to propositions derived from theory, claims of statistical significance of a regression result, or a bald assertion. Consider: "Why do people still ranch? And why, lately, have more ranchers started to write poetry about their lives?" (*New York Times,* July 5, 1992). If you were to make such a poetical claim, you could expect to be asked for supporting data. But, to take an example of the other kind, after you quote World Bank GDP tables to show real income growth in developing countries slowing up in the 1980s, your assertion that real income growth slowed in the 1980s is likely to be unchallenged.

This is wrong, for the supine acceptance of data can cause scientific mischief of the worst kind—believing we know things that aren't so. The purposes of this paper are to show how the data agricultural economists use can mislead, and to discuss what might be done about it.

How Data Can Mislead

Morgenstern, in his classic work on economic statistics, found much to criticize in U.S. official statistics. He emphasized the difficulties of obtaining unbiased information from questionnaires, inevitable recording and computational errors, and problems caused by using "benchmarks" and revisions of previously published data. Bonnen placed more emphasis on conceptual obsolescence in the agricultural data system, particularly reliance on the farm as the basic unit of observation. More recently, studies have focused on technical problems in aggregation, over commodities (Ball) and over farms (Chambers and Pope).

This paper focuses on a problem area different from Morgenstern's, Bonnen's, or aggregation issues; namely, the consequences of creating data in cases where the primary statistical base is insufficient. Economic theory has provided ways to extrapolate from the primary statistics gathered in surveys to a representation of the facts we want. This process generates theory-laden data that are often ingeniously constructed but also are often dubiously constructed. Most insidious are cases in which the data generated are acceptable for some purposes but disastrously inappropriate for others. Unfortunately, the inappropriate uses tend to be ones in which economists are most interested for their scientific work (testing economic hypothesis using historical data). Examples follow.

The M-W milk price. Each month the National Agricultural Statistics Service (NASS) publishes the Minnesota-Wisconsin (M-W) price. The M-W price is an estimate of the average price paid for grade B milk by plants that manufacture dairy products. Currently, 67 plants are surveyed each month, and asked what price they are paying farmers for milk during the current month and what price they paid the previous month. The data-generating process is as simple and direct as for any statistics published by the U.S. government.

The M-W price has been criticized on the ground that it is not representative of U.S. prices paid for milk used in manufacturing (U.S. General Accounting Office), but beyond the issue of representativeness, the time series behavior of the M-W price could mislead economic researchers studying milk markets. To achieve timeliness, NASS surveys the M-W plants be-

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Fellow's lecture.

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fore the plants have paid producers for the full month. Indeed, some of the plants have not paid farmers even for the first half of the month and must estimate what they will pay. So we do not have a report of actual market transactions.

The plants' estimates will turn out wrong when market conditions change. NASS statisticians observe how the 67-plant M-W price has recently differed from the final price (obtained from a larger sample of plants, in the following month), and adjust the M-W price. Thus, in two respects the published M-W price estimate is not an average of prices actually paid for milk: plants forecast their prices, and NASS statisticians attempt to correct for under or over-shooting this may cause. There is no fixed rule for making these adjustments. It is likely that the NASS statisticians will follow an error-correction procedure; if recent published M-W prices exceed the price actually paid during the previous month, then the tendency will be to reduce the plant-reported price.

These procedures are probably close to the best attainable given the requirement of timeliness, but the resulting data could easily lead one of our profession's ever more sophisticated researchers astray. A natural question to ask about the grade B milk market is whether it is an efficient market. One might test for efficiency using time series econometrics. Do milk prices follow a random walk? If we fit an ARIMA model, what we will find is not a random walk but a moving-average process, placed there by the NASS error correction procedure. So the researcher can reject efficiency—finding there is information in last month's M-W price that enables one to predict this month's M-W price better than predicting that this month's price will equal last month's (seasonally adjusted) price. But this result occurs because NASS published a price chosen partly by looking at the pattern of past prices.

I call the study of how primary statistical information is made into economic data "factology." The neglect of factology risks scientific ruin. And factology can be difficult. A researcher could not easily find out the pitfall in the data. In an obscure source, NASS (1992) discusses their use of additional information beyond the M-W survey results, but nowhere does NASS publish information that describes how the plant-reported data are adjusted. One could avoid trouble by using the revised data, but it makes sense to use the originally published data because these data, not the month-later update, establish the M-W price used in federal orders.

Feed use. USDA publishes data on feed use of corn and other grains. However, there are no survey data on the quantity of grain that farmers feed to livestock. NASS has survey data on production, exports, and stocks at the beginning and end of the year. Analysts then take production plus beginning stocks minus exports minus ending stocks as an estimate of domestic use; and using estimates (not from random sampling) of commercial use in non-food manufactured products like ethanol, feed use is the residual left over. This is the most efficient way to obtain a feed estimate, but nonetheless these data are constructed rather than observed. All the non-sampling and sampling errors in the grain production, stocks, export, and commercial use data are summed in the feed figures.

The residual nature of the feed data can be important for econometric work, since an errors-in-variables problem is inevitable if one attempts to estimate a price-dependent demand function for grain fed, which many have done. The tendency is to observe fluctuations in "feed" not correlated with price changes and hence to overstate the elasticity of demand for feed. To its credit USDA, in its *Agricultural Outlook* and *Feed Situation and Outlook,* clearly labels these data as "feed and residual," so the econometrician is warned.

Crop forecasting. Bradford and Kelejian (1978) carried out eye-opening research on the value of crop forecasting information. One of their main insights is unusable in absence of factology. They point out that when a series of monthly crop yield estimates is made as the growing season progresses, the optimal estimate of the true potential yield[1] is *not* the latest estimate. This is so despite the fact that the latest estimate is made with the most information available, and hence with smaller expected error than earlier estimates. The reason is that even the latest and best monthly estimate still contains measurement er-

[1] The term "potential yield" is a reminder that, contrary to common belief, the NASS estimate is *not* the statistically expected value of yield given current information. Instead, the estimate assumes normal growing conditions in the remainder of the growing season. Thus, the NASS early-season orange crop estimate does not incorporate the probability of crop losses due to frost. This approach makes a difference in grains in a year like 1992, when the corn crop got a late start and the downside risk from an early end to the growing season is not negligible. NASS takes no account of this risk. Thus, in August 1992 the NASS corn yield estimate exceeds the statistically expected corn yield, even if it accurately measures potential yield as of August.

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ror. Consequently, earlier forecasts, which can be viewed as independent estimates of the same unknown parameter (yield), contain additional information. Only a "naive" forecaster (Bradford and Kelejian's term) would use the latest estimate as a crop forecast. A "sophisticated" forecaster would construct a weighted average of all NASS's earlier estimates. The weights depend on the parameters of the joint distribution of the monthly estimates—essentially like a GLS procedure in econometric estimation.

The problem with this approach is that NASS itself is not naive. It does not publish an independent objective yield estimate each month. Each state reports three indicators of the state's yield to Washington after a farm survey at the beginning of each month: farmers' assessments of general growing conditions in their area, farmers' assessments of their own yields, and the NASS enumerators' objective yield measurement (when available). A NASS board in Washington then assesses all the indicators of yield, including the estimates of a month earlier. This is not done using a pre-specified formula—in which case a computer could replace the NASS Board—but through a consensus of the Board members based on their experience and the full information before them. If the current estimate is substantially different from earlier estimates there is a tendency not to adjust the prior NASS estimates fully to rely only on the current month's information. Thus, the NASS estimate already does with its prior information what Bradford and Kelejian recommend doing with published NASS estimates, and it is a tricky business to apply the sophisticated approach to the published NASS data.

The publication describing USDA crop and livestock estimates outlines NASS procedures (USDA, 1989, p. 9 and p. 12), but the role of earlier estimates in formulating the current month's estimate is not discussed. Nor do the monthly publications that report the crop estimates describe the estimating procedure. By comparing the current month's with earlier estimates, the monthly publications may indeed encourage the reader of Bradford and Kelejian to use the sophisticated approach with NASS data. This example reenforces the point that intelligent use of USDA data requires a full and detailed disclosure of how the data are constructed.

Investment in farm buildings. An instance where USDA data have had a more explicit theoretical content is the statistics of investment in farm structures—farm buildings such as barns and grain storage facilities. The USDA procedures as of the early 1970's were criticized by Bhatia (1971). No direct survey data on expenditures for farm structures were available, except in a one-time 1955 survey. The 1955 data were used to estimate the relationship,

$$(1) \qquad e_i - e_j = 4.8276\,(s_i - s_j)$$

where i and j represent farms in different gross income (sales) classes, e is expenditures on structures, and s is gross income. Equation (1) was used to move the aggregate U.S. level, $E_t(= \Sigma_i\, e_{it})$, where t = year, from 1955 to prior and succeeding years by the relationship

$$(2) \qquad E_t - E_{t-1} = 4.8276(S_{t-1} - S_{t-2}).$$

The conceptual basis for using the cross-sectional result over time, the use of the lagged change in S, and the precision of the parameter value are all dubious, as Bhatia points out.

The point I wish to emphasize here, however, is that even if the procedure were unexceptionable, the resulting data would be misleading if labeled (as they are) as a time series of "farm gross capital expenditures" with no indication of how they were constructed. In recent years, survey data have been available to provide better annual information, but the historical data are still labeled as "expenditures" on "service structures and land improvements." An explanatory footnote states: "Includes service buildings, other structures, fences, windmills, wells, dams and ponds, terraces, drainage ditches, tile lines, other soil conservation facilities, and dwellings not occupied by farm operators" (USDA, Nov. 1991, p. 42). Rather than providing a needed warning, this footnote's deceptive specificity would deflect the user from recognition that for more than half of the 1945–1990 annual figures, the year-to year changes are just the movements of lagged U.S. aggregate sales.

Where this situation could lead to disaster is in an attempt to use these "capital expenditure" data to test, say, alternative models of investment. Indeed a notable exchange between Grove and Scott and Heady (1967,1969) turns on this very issue. Scott and Heady studied investment in farm buildings using the USDA data on expenditures on farm buildings. They considered, among other hypotheses, the "accelerator" model in which investment is a lagged function of output, and the neoclassical model in which the interest rate is a primary determinant of invest-

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ment. Of course, as Grove knew (but Scott and Heady did not) the data were actually constructed by U.S.D.A. according to a version of the accelerator model. So the significance of the lagged gross income variable is explained by factology, but not by the facts.[2]

Agricultural productivity. In productivity measurement the data themselves are interpreted as telling a story. The USDA total factor productivity index has continued to rise at a 1–2% annual rate in the 1980's when non-farm productivity indexes were rising more slowly or even declining. This is seen to speak well for the efficiency of our farmers and the success of U.S. agricultural research. But are the productivity data a description of facts?

It is helpful to think of a change in total factor productivity as a measure of technical change, and the level of a productivity index as corresponding to parameter in a production function (see Chambers, chapter 6). Thus, if firms have identical Cobb-Douglas production functions, minimize costs, and technical change is Hicks neutral, we can express output y with changing technology at different times, t, as

$$y_t = A(t) \prod_{i=1}^{n} x_{it}^{\alpha_i} \tag{3}$$

where x_i is the quantity of the ith of n inputs, α_i is its constant cost share, and $A(t)$ is an efficiency indicator for the technology at t. Then an appropriate measure of productivity is

$$A(t) = \frac{y_t}{\prod_{i=1}^{n} x_{it}^{\alpha_i}} \tag{4}$$

Thus, the ratio of output to an appropriately specified input index measures both productivity and the technology parameter of the production function.

A major accomplishment of analytical work in productivity measurement is to find the best productivity indicator for more general representations of technology. Ball and Antle and Capalbo recommend measuring changes in total factor productivity (TFP) from time $t-1$ to t as

$$\ln\left(\frac{TFP_t}{TFP_{t-1}}\right) = \frac{1}{2}\sum_{i=1}^{m}(r_{it} + r_{i,t-1})\ln\left(\frac{y_{it}}{y_{i,t-1}}\right) - \frac{1}{2}\sum_{j=1}^{n}(s_{jt} + s_{j,t-1})\ln\left(\frac{x_{jt}}{x_{j,t-1}}\right) \tag{5}$$

where the y_i are output quantities for each of m products, x_j are input quantities for each of n inputs, r_i are output revenue shares, and s_j are input cost shares. The AAEA Task Force on Productivity Measurement showed that appropriate weighting and aggregation can make a substantial difference in measured productivity. But even bigger problems are created by the use of economic theory to fill gaps in the data.

Capital services. The practical reasons for moving beyond the straightforward partial productivity indicators (output per acre, output per farm worker) are that labor and land together account for less than half of the cost of producing U.S. farm output, and the share of other inputs is rising over time. The most important of these other inputs is physical capital (Ball, 1985, tables 8 and 9). Yet the problems of measuring the capital inputs are so severe that one might despair of obtaining a meaningful measure.

Tractors and other farm machinery make up the largest fraction of capital inputs. The 1987 Census of Agriculture asked "What is the estimated market value of ALL machinery, equipment, and implements usually kept on this place and used for the farm or ranch business?" (U.S. Department of Commerce, 1989, p. D-5). Farmers' answers summed to $85.8 billion for the United States. This gives a good base to work from in estimating capital inputs. The problem is that the previous occasion when (approximately) this question was asked was in the 1945 Census.[3] Where do we get data for the intervening years? USDA constructs them.

In recent years, the Farm Costs and Returns Survey provides an estimate of expenditures by farmers on machinery and equipment. Before 1985 USDA relied on analysts' informal sur-

[2] A more recent study (Hubbard and Kashyap, 1992) investigates the effect of farms' net worth on investment in farm equipment, in particular whether capital rationing due to asymmetric information between borrowers and lenders explains investment. The early years of their 1910–86 investment data are in part constructed from data that may create similar spurious relationships. The data source is given as U.S. Department of Commerce, but these are really the same old USDA data (see footnote 9 below).

[3] The 1945 Census enumerator instructions were to fill a blank entitled "value of farm implements and machinery, including trucks and tractors (omit automobiles)." (U.S. Department of Commerce, 1945).

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veys of the machinery and equipment industry to obtain an estimate of annual sales. From the 1945 benchmark, estimated new purchases are added to the stock and estimated depreciation is subtracted out. (For details, see USDA 1989.)

Tostlebe investigated the ratio of the capital stock to output and to labor. But it has since become accepted that the appropriate input measure is not the capital stock itself but a service flow—the quantity of services observed in a market transaction such as custom harvesting. Unfortunately, the quantity of services (as distinct from the stock) is not observable. Even the most economically well trained farmer would have difficulty reporting the quantity of services from the use of a combine—the response would be either the availability of 1 combine (say) or the number of acres harvested. So there is no alternative to having analysts construct an annual service flow. USDA measures this as the sum of estimated depreciation (the amount of the stock used up), repairs and maintenance, and the opportunity cost of capital. The last item means that USDA must use an interest rate, for which they have chosen the 1976–78 interest rate on farm real estate debt (USDA 1989, p. 12).

Using the resulting capital input quantity measure could cause problems for an econometric investigation of agricultural technology. The service flow from buildings, in the historical data, still uses data derived in part from the constructed "expenditures" discussed earlier. The depreciation of capital is dubiously measured (Penson, Hughes, and Nelson), by depreciating equipment at a geometric rate that implies almost-new equipment is much less productive than engineering information suggests is appropriate.[4] Generally, the procedures place substantial inertia in capital services, which one might not see if actual service flows such as custom harvesting services could be observed. An investigator might find inflexibility in the data that doesn't exist in fact.

From the perspective of the story that productivity data tell, a comparison of the capital stock data that we have with USDA's capital service flow raises doubts. Recent Agriculture Censuses have asked about numbers of the main machinery items: tractors, trucks, combines, and forage harvesting equipment. (See USDA 1991, table 31, for these data.) The simple sum of these items, or any price-weighted index of them, increased between 1978 and 1987. Tractors are the only item that declined, from 4.63 million in 1978 to 4.61 in 1987 (and this is probably offset by the 1987 tractors being bigger). Yet the USDA mechanical power and machinery index declined 30% between 1978 and 1987. This probably results from declining new equipment purchases coupled with the fast depreciation used, so that the 1987 stock, while bigger, is older. But are the productive services this equipment provides that much smaller? This is doubtful, although the expansion of reduced-tillage production methods probably means that tractors are sitting idle for more hours each year. If the understatement of 1987 relative to 1978 capital inputs is as large as it seems, the result would be a significant overstatement of the rate of productivity growth in the 1980s.

Labor inputs. Another potentially serious problem for researchers using USDA input and output data is USDA's former practice of using labor "requirements" with output and acreage data to construct the hours of farmwork for each year. This practice has now been discontinued and replaced by data from the Bureau of Labor Statistics and the Farm Cost and Returns Survey (USDA 1989, p. 9). But anyone using the 1910–90 published data to investigate production relationships could easily fall into a trap because the 1910–46 data still are those created by the old method. For example, someone looking for structural change in agricultural production functions could easily find more substitutability between labor and land after 1946 than before simply because the pre-1946 data impose short-run fixed coefficients.

Other problems with productivity data depend on the uses to which the data are to be put. Researchers in production economics naturally use total factor productivity as a measure of the efficiency parameter $A(t)$ in equations (3) and (4). Yet some USDA procedures are inconsistent with this interpretation.

[4] Depreciation of farm buildings is even more problematical. It is assumed to be 2 percent annually. The AAEA Task Force (1980, p. 32) objected to this figure as being too low. The analysts responsible, in the ERS Inputs, Technology, and Productivity Branch (ITPB) defended as follows:

> The 2-percent depreciation rate is lower than that used in the farm income estimates of ERS, since ITPB analysts felt that estimates of depreciation plus repairs on buildings were too large, due to overreporting of maintenance expenditures. A lower depreciation rate compensated for any resulting measurement error. (USDA, 1989, p. 11).

This assertion has nothing to recommend it but boldness. It is certainly far removed from the world of sampling errors that we like to pretend are the main problems we have to deal with in obtaining econometric results from data.

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USDA's quantity index of farm real estate includes not only cropland planted but also the acreage idled under government programs. Idled cropland has averaged about 60 million acres in recent years or about 20 percent of cropland harvested. Thus, the real estate input substantially overstates land used to produce U.S. farm output, and understates productivity in the sense of the efficiency of U.S. farm production.

A counter argument is that the idled land could be producing output; and if policies cause the output not to be forthcoming, that inefficiency is properly incorporated in measured U.S. agricultural productivity. A similar argument is that in aggregating inputs or output, international or other appropriate shadow prices should be used if policies distort U.S. prices.[5] In both of these instances, the recommendation makes sense if we want a productivity measure that provides a broad indicator of efficiency—how actual output compares with the output that could be achieved under alternative policies. But we are no longer measuring $A(t)$ in any meaningful sense, and there is a price to be paid in moving to such a broad measure; namely, the rigorous economic arguments that led to equation (5) as an appropriate approximate index no longer hold, and the whole measurement issue has to be rethought.[6]

A similar problem arises with USDA's inclusion of insurance purchased as a category of miscellaneous inputs. Insurance does not add to output through obtaining more output from given inputs (although insurance will likely cause farmers to change their input use, and hence influence output). The USDA productivity index is in this sense also a broad measure, measuring the effects of both farm production and utility functions, as opposed to narrowly measuring shifts in $A(t)$.

There is no "right answer" to the choice of a broad or narrow conception of productivity. For data that themselves tell a story, and a different story depending on the use to which the data are put, it is especially important for the user to undertake an investigation in factology before telling the story.

Farm income. Data on farm income provide another instance where the figures themselves are taken to tell a story, and indeed have direct political implications. It is therefore especially important that these data do not mislead.

A big problem with farm income data is conceptual ambiguity, stemming once again from the different uses to which the data are put. Particularly important are two uses of farm income data: as a measure of well-being of farm people, and as a measure of returns to farming. USDA data are constructed to be able to serve both purposes. This has led to misunderstandings in which farmers and economists talk past one another because they have different conceptions of income in mind.

For both purposes, it is reasonable to start with farm commodity receipts and subtract farmers' production expenses, including capital inputs. Economists are inclined to think of the data as a measure of the farm household's multiyear budget constraint. Since farmers are self-employed, this means that goods and services produced on the farm and consumed by the farm household are part of farm income; and it means off-farm income earned by farm household members should be added. The multiyear aspect means choices and events this year that increase future income should be taken into account—a farm household's income this year is current consumption plus savings. And savings include any increase in wealth, notably real increases in the value of assets owned by the farm household.

Farmers tend to see farm income as the returns to their effort and investment in farming. From this perspective, off-farm earnings should be excluded; and the rental value of the farmer's house should be excluded.

USDA deals with these differing ideas reasonably, by offering a variety of data from which data users can construct the measure desired. The *National Financial Summary* for 1990 reports the following measures (USDA 1991, p. 10):

	billion $
net farm income	50.8
net cash income	61.8
net business income of farm	46.0
returns to farm operators	49.4
off-farm income of farm operator households	67.0

[5] For example, the AAEA Task Force argued that measuring the rental value of Federally owned grazing land by using the legislatively established grazing fees understates the value of grazing services; they recommended that a market grazing rental fee for comparable private land be used instead (AAEA, p. 46).

[6] It is also true that distortionary policies cause a necessary rethinking of the narrow measure of productivity in terms of $A(t)$. If we attempt to "improve" the USDA procedures by subtracting idled acreage from the real estate quantity index, two problems must be faced: (1) idled land is selected by farmers to be low-quality land, and (2) idled land does produce output in the form of some grazing and increased yield when the land is brought back into production. So the introduction of a land-idling program would increase (narrowly) measured productivity, even if $A(t)$ did not change.

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Nonfarm personal income was $4,574 million in 1990 (U.S. Council of Economic Advisers, 1991, p. 315). The most straightforward measure for comparing farm and nonfarm income is per capita income of the two groups. In 1990 the U.S. nonfarm population was 247 million, so per capita income is $4,574/247 = $18.5 thousand. What is the comparable farm income measure? The USDA data permit us to add any of the farm measures to off-farm income and divide by the estimated 4.7 million farm population. Using net farm income we obtain (50.8 + 67.0)/4.7 = $25 thousand. Thus, the data say farm people on average are better off than nonfarm people by 35%. But is this comparison legitimate?

The following have been cited as reasons why such a comparison overstates farmers' well-being as compared to nonfarm people:

—farm income includes the rental of value of owned housing, nonfarm income does not;
—farm income includes home consumption of items produced by the household, nonfarm income does not;[7]
—farm income includes the value of inventory accumulated, nonfarm income does not;
—farmers work more hours than non-farmers;
—farmers have more capital invested than nonfarmers.

The following have been cited as reasons why the comparison understates farmers' well-being as compared to nonfarm people:

—the farm population includes many low-income people who live on small farms, yet get almost all their income off-farm; and excludes well-off farm owners who do not live on their farms. The population of commercial-scale farm operators would show higher average income;
—the data subtract out farm property taxes but not nonfarmers' property taxes;
—the data are pre-income-tax, and farmers pay less income tax than nonfarmers per dollar earned;
—there are more people per household on farms, so income per household would show farmers in a more favorable income position relative to nonfarmers;
—nonpecuniary benefits (such as clean air) are higher in rural areas.

These points lead in a number of directions and are too complicated to follow up on here. But one needs to have considered them all carefully before making a judgment on the "facts" of farm/nonfarm income comparison.[8] USDA has become circumspect about making such comparisons. The 1991 publications in the Economic Indicators of the Farm Sector series omit any farm/nonfarm income comparisons.

Exports and imports. Trade statistics are interesting for factology because each commodity shipment is both an export and an import, and so is subject to being counted twice, by two different countries. This means that empirical investigation of data quality is possible, a task well undertaken in Hiemstra and Mackie (1986). A notable practical finding is that U.S. grain exports to Canada are more accurately estimated by Canadian import statistics than by U.S. export statistics.

Economic theory and factology. Such empirical investigation can give sustenance to a theory of factology—explaining why some data are more accurate than others. A hypothesis is that regulation leads to better data. Application: since imports are more often taxed than exports, governments collect more detailed data on imports. A more refined theory should relate data to the incentives created by the regulation. Application: the more heavily imports are taxed, the more data errors are created by smuggling. Immigration and imports of cocaine are so heavily regulated that our statistics are suspect. On the other hand, data on subsidized activities are liable to be overstated; for example, the extent of conservation practices on set-aside or Cropland Reserve Program acreage. Federal income tax data provide both incentives—they provide a lower bound on agricultural sales of filers, and an upper bound on deductible expenditures, hence understating net farm income.

[7] "Noncash income" measured by USDA includes housing services, the farm value of food consumed on the farm, and (after 1985) firewood (USDA 1988, pp. 13–14). It excludes the value of canning and other food processing services and clothing produced in the household—in short, the noncash income traditionally generated by farm wives.

[8] The trend over time in comparisons of income may mean more than their level. For example, the same measure of per capita income that gave farmers 135 percent of nonfarmers' incomes in 1990 give farmers 53 percent of nonfarmers' incomes in 1960 (USDA, 1991, p. 11; U.S. Council of Economic Advisers, 1991, p. 315).

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What Can Be Done?

The pitfalls of data lead us into two dangerous temptations as alternatives to using published data.

Drawing Conclusions from Unquantified Observation

It is tempting to believe that an experienced economic observer can sort out the essential causes and effects of events using direct observation through careful attention to economic activity as it occurs, without any formal economic theorizing or statistical work. "Fact-finding" missions are sometimes sent expressly to undertake such efforts. Visiting the former USSR, for example one may be inclined to assert that Russia is having great difficulty getting off the ground economically because its history provided too little experience in entrepreneurial investing and too much in coping with severe bureaucratic constraints. No data are available to test the idea formally, but observation and reported events make it plausible.

This intuitive approach is a useful starting point for economic inquiry, but it's a sad day for our scientific aspirations if we stop at this point and write up our results. The problem becomes apparent when experts disagree. We end up settling disputes by force of rhetoric or debating skills—or, more usually, not settling disputes.

Stylized Facts and Empty Models

One step up from unquantified generalization is the hardening of unquantified observations into "stylized facts"—facts put forth as assumptions without direct support from data. Inferences from stylized facts can be interesting, but the results have no more empirical force than the stylized facts; and there may be no more support for these "facts" than for conclusions from unquantified observation.

It is just as well to call assumptions assumptions. The revealed pitfall in economists' efforts to model agriculture on the basis of assumptions with supposed empirical content is that one too often reaches erroneous conclusions. Schultz's *Transforming Traditional Agriculture* is in large part devoted to showing where even well-received models of economic growth and of agricultural development went wrong, and how scientifically serious the adverse consequences were.

In short, there is no substitute for working with data. How can they be improved? Bonnen states: "The difficulty with most of the literature on the problems of agricultural data and information systems is that the authors often overlook the forest and see only the trees." (p. 402). I plead guilty with an explanation. I have avoided big-picture conceptual or data-system issues because others' attempts have been useless and I can't do better. Also, I believe the big problems are with the trees.

Brandow et al. tried a big-picture conceptual recommendation:

> For purposes of measuring income in the farming sector, we recommend moving completely to an "establishment basis" in preference to a products basis. This means that ancillary products and services produced by farm establishments, particularly those described by SIC code 07 (agricultural services) should be included in gross receipts or sales. (p. 18).

On the other hand, the AAEA Task Force on Productivity Measurement states:

> We should concentrate on the product, rather than attempting to define the productivity of particular social entities or business establishments. Our underlying concern is how the function of producing raw agricultural products is being performed. Concerns as to which firms perform this function and the labels attached to these firms are of low priority. (p. 43).

These contradictory recommendations derive from different uses of data. In measuring farm income, Brandow et al. believed that counting large corporate broiler producers, for example, as farms would obscure more than it would illuminate. But for purposes of productivity measurement, the focus is properly on the output-producing process rather than who does the producing. (Although as long as farm policy is specified on the basis of products rather than people, a product-based income measure is more germane than one might expect.) Here, as is often the case, the big-picture advice offered to USDA adds up to less than the sum of its parts.

The main source of progress in data in recent years have focussed on the trees, not the forest, particularly the efforts of the National Agricultural Statistics Service (NASS), the Economic Research Services (ERS), and the World Agricultural Outlook Board of USDA. The overall trend can be seen most clearly in the NASS budget. Using the GNP deflator to express real val-

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ues (in 1987 dollars), the NASS budget fell by 15 percent from $75.3 million in Fiscal 1980 to $64.3 million in Fiscal 1983. The retrenchment at this time eliminated about 30 statistical reports on commodities and farm labor, and reduced the frequency and quality of many series in other reports. Since that time, the NASS budget has been restored and increased in real terms to $76.1 million (1987 dollars) in Fiscal 1992. The expansion has enabled a substantial increase in statistical activity on agricultural inputs, particularly pesticide use by farmers.

With respect to economic statistics, USDA's Farm Costs and Returns Survey is a major, and very costly, investment that has greatly improved the primary data base for farm costs, farm income, and productivity measurement. This effort will reap dividends for many years to come. Even with these data, many details about farm operations do not have a solid factual basis. With respect to operator labor, farmers are expected to recall their monthly labor hours from over a year earlier. They are asked to value assets at a particular time over a year in the past. Errors of the types discussed by Morgenstern are sure to be present. Nonetheless, this data project is by far the best effort going or in prospect on farm data. The leadership and dedicated experts in NASS and ERS are in my view heroes in the struggle for better data. The profession is fortunate to have this resource because an ongoing primary economic data program is beyond the financial reach of Colleges of Agriculture today.

Despite the progress made, important issues in factology remain in an unsatisfactory state. USDA still has far to go in describing in detail how its published data are constructed. For intelligent use of these data in econometric work, the details are crucial. For example, Hauver and Ball (1991) describe and present data for a new total factor productivity index. They made great strides in meaningful aggregation, but are still stuck with real estate data before 1970 that are based in part on farm sales, as discussed earlier.[9] And the reader still gets no clue how these underlying data were constructed. Econometric land mines remain in place.

[9] The historical data for the new indexes of investment in farm structures comes from the U.S. Department of Commerce. But the source statement for the Commerce data is: "The farms industry investment estimates were available from the U.S. Department of Agriculture for the years beginning in 1910." (U.S. Department of Commerce, 1987, p. xiii). These are the same data whose problems were discussed earlier! Moreover, Commerce "extrapolated" the highly dubious USDA data prior to 1947 to estimate investment in agricultural services, forestry and fisheries.

Conclusion

Agricultural economists and other social scientists tend to take data as facts. The facts these data are meant to establish are important facts—the price of food, the well-being of farmers, their costs of production, the productivity of agriculture, etc. The problem is that data are not facts. Facts are what is really there. Data are quantitative representation of facts, which statistical workers and economists concoct.

It was recently written of J. M. Keynes:

> Like many theorists, Keynes was hostile to the notion that an econometric model could be a source of new economic ideas. In his view, empirical work did not lead to acts of discovery in economics. It could only illustrate a priori theory. If an empirical model contradicts the theory, the data are wrong—not the theory. (Heckman, p. 878).

This view looks dubious if not crazy. But the last sentence can be modified to state an important truth: whenever we test a hypothesis statistically we are testing the data as well as the hypothesis.

Data are important because they are the raw material for economic investigations and because the data sometimes tell a story by themselves. Data are raw material but not natural resources—we create the data, and this is the source of problems. This paper presented an album of problems, seeking to illustrate two main types of pitfall: data causing useless or misleading results of economic investigation; and data telling an untrue or misleading story. The former pitfall is especially dangerous for workers on the frontiers of economic research—looking for subtle facts like efficiency in short term commodity price movements, or rigidity in investment or supply adjustment processes. Short-term movements are particularly likely to be the creations of statistical workers as opposed to recorded observations of market phenomena. The second problem is especially serious for policy workers because, typically lacking applicable results from econometric research, they rely on the data speaking for themselves. But the key statistics with stories to tell—farm income, cost of production, productivity growth—are so much a matter of construction that the user is highly at risk here also.

The bottom line is, for data producers, that full disclosure of procedures and clear labelling of data series are essential; and for data users, be careful and investigate the data before using them as, for example, in Nerlove 1958. I have

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eschewed wide-scope conclusions that the public should spend more on agricultural data or that data manufacturing should be organized and conducted in new ways or use new concepts.

The problems surveyed in this paper have been limited to U.S. farm data. In food marketing, other data issues are equally troubling, e.g., measuring price spreads. Data problems appear to be even more serious outside the United States. So perhaps this discussion has been too sanguine in tone.

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